## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. $(2;-1)\cdot(1;2)$ ?

2. Does the point (1; 2) satisfy 2 $x-y=0$ ?

3. Solve $x+y=5,\ x-y=1.$

4. Distance from (1; 2) to (4; 6) ?

5. Complete $x^2-4x.$

6. Solve $x^2+y^2$ = 0 in $\mathbb{R}^2.$

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Present the two conditions characterising a projection. Give a proof geometric proof and an algebraic proof of a perpendicular bisector.

## Investigate before formalising

From $P(4; 3)$ and $d : y = 2x$, compare the distances to three points of d, then find and prove which one is shorter.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.